# Practice workbook · Match human cell structures to their functions

Naming an organelle from its job, or giving the job of a named organelle, including nuclear membrane, ER and mitochondria in context.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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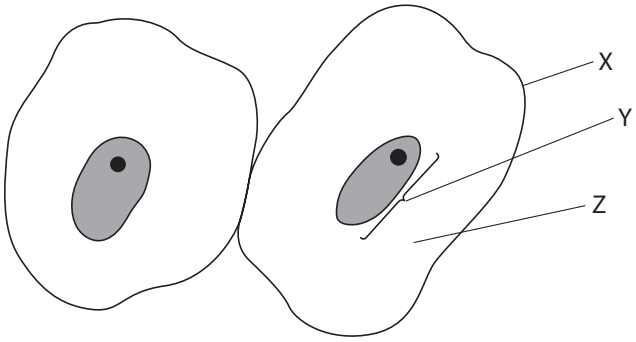
Type ID: HBI-MATCH-ORGANELLE-TO-FUNCTION

# 1. Question 7 · part a · 6 marks

June 2025 Paper 1R

Worked-example retry · Attempt part a only.

**7** The diagram shows two cheek cells from the inside of the mouth.




(a) Complete the table by giving the missing information.

**(6)**

| Part | Name of part | Function of part |
|---|---|---|
| X | | |
| Y | | |
| Z | | |

(b) Describe how a student would obtain and prepare these cells so the student can view them using a light microscope.

**(4)**

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## 1. Question 7 · part a · 6 marks · Working and answer

Working and answer

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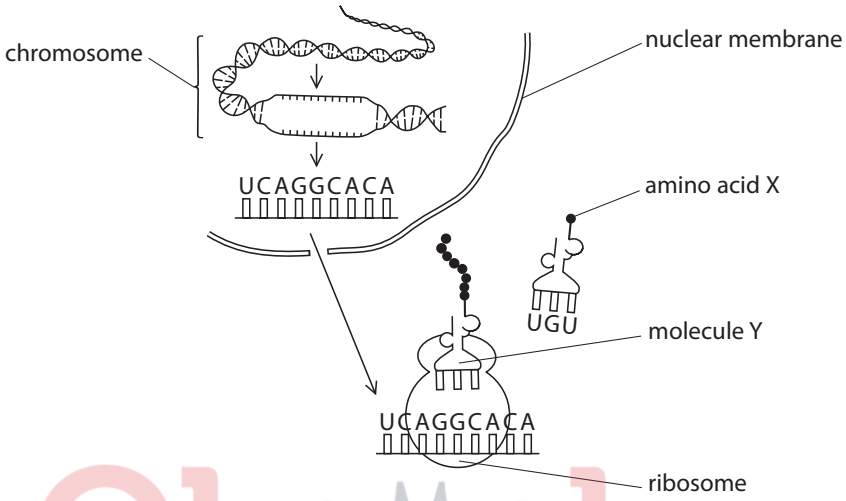
## 2. Question 6 · part a · 2 marks

June 2019 Paper 1R

Further practice · Attempt part a only.

**6** Protein synthesis has a number of stages.

The diagram shows some of the stages.




The table lists the base triplets of DNA that code for some of the amino acids found in a protein.

| Base triplet of DNA | Amino acid |
|---|---|
| AGT | serine |
| CCG | glycine |
| TGT | threonine |
| GTA | histidine |
| CAA | valine |
| TCC | argenine |
| ACA | cysteine |

(a) Describe the function of the nuclear membrane.

**(2)**

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## 2. Question 6 · part a · 2 marks

June 2019 Paper 1R

Further practice · Attempt part a only.

(b) Which is the correct sequence for the three unpaired bases on the tRNA molecule labelled Y?

**(1)**

- ☒ **A** AAU
- ☒ **B** CCG
- ☒ **C** GGC
- ☒ **D** UUA

(c) Give the name of amino acid X.

**(1)**

(d) Describe the process occurring on the ribosome.

**(6)**

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## 2. Question 6 · part a · 2 marks · Working and answer

Working and answer

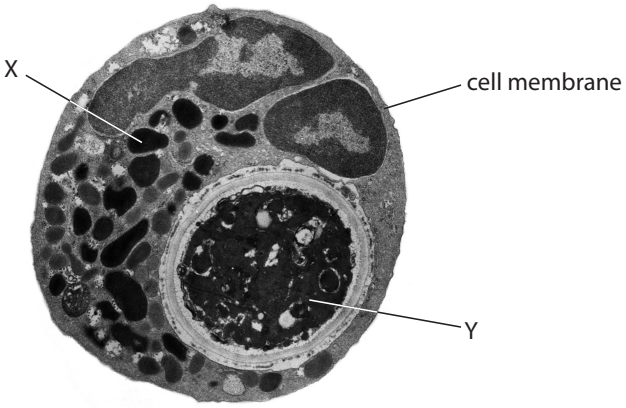
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# 3. Question 2 · part a.i · 2 marks

June 2019 Paper 2

Further practice · Attempt part a.i only.

**2** (a) The image is an electron micrograph of a human body cell.




(Source: © Science History Images/Alamy)

(i) The table gives information about parts of the human body cell.

Complete the table by giving the missing information.

**(2)**

| Part | Name of part | Function |
|---|---|---|
| X | | release energy from glucose |
| Y | nucleus | |

(ii) Electron microscopes and light microscopes can both be used to view body cells.

Which of these is an advantage of using an electron microscope?

**(1)**

☒ **A** more cell structures can be seen with greater resolution

☒ **B** more cell structures can be seen with less resolution

☒ **C** fewer cell structures can be seen with greater resolution

☒ **D** fewer cell structures can be seen with less resolution

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## 3. Question 2 · part a.i · 2 marks · Working and answer

Working and answer

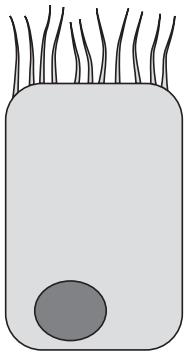
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# 4. Question 2 · part a.ii · 3 marks

November 2020 Paper 1R

Further practice · Attempt part a.ii only.

**2** The diagram shows a cell from the human breathing system.



(a) (i) Add these labels to the diagram.

- cilia
- cell membrane
- nucleus

**(3)**

(ii) Each of the cell structures has a particular function.

The boxes give a list of structures and functions.

Draw one straight line from each structure to its function.

**(3)**

| Structure | Function |
|---|---|
| cilia | holds the genetic code |
| cell membrane | site where most chemical reactions occur |
| nucleus | controls the entry and exit of substances |
| | makes proteins |
| | moves mucus through the trachea |

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# 4. Question 2 · part a.ii · 3 marks

November 2020 Paper 1R

Further practice · Attempt part a.ii only.

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(b) (i) Cigarette smoking affects the function of the cell shown in the diagram.

Name a substance in cigarette smoke that causes damage to this cell.

**(1)**

(ii) Describe how cigarette smoking affects the cell shown in the diagram.

**(2)**

**(Total for Question 2 = 9 marks)**

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Working and answer

# 5. Question 8 · part a.iii · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a.iii only.

**8** (a) CFTR is a protein that is found in cell membranes.

This protein is responsible for transporting sodium ions across cell membranes using active transport.

(i) Describe the process of active transport.

**(2)**

(ii) In some people a DNA mutation causes the production of a faulty CFTR protein.

This results in cystic fibrosis.

Explain how an individual can inherit cystic fibrosis from their parents.

**(2)**

(iii) The faulty CFTR protein is unable to bind to the endoplasmic reticulum in the cell following protein synthesis.

Suggest how this might affect the CFTR protein.

**(2)**

(iv) The CFTR protein is made up of 1408 amino acids.

Calculate the number of nucleotides found in one strand of the CFTR gene.

**(2)**

number of nucleotides = .....

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## 5. Question 8 · part a.iii · 2 marks · Working and answer

Working and answer

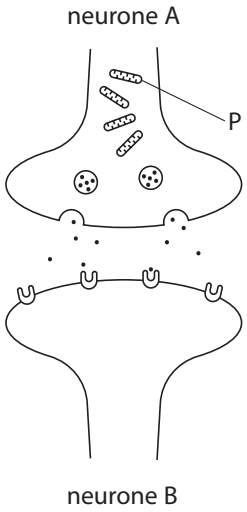
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# 6. Question 8 · part a.ii · 3 marks

November 2020 Paper 2R

Further practice · Attempt part a.ii only.

**8** The diagram shows the gap between two neurones, A and B.




(a) (i) Draw an arrow on the diagram to show the direction of travel of an impulse.

**(1)**

(ii) Explain the function of structure P in the transmission of the impulse across the gap.

**(3)**

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## 6. Question 8 · part a.ii · 3 marks

November 2020 Paper 2R

Further practice · Attempt part a.ii only.

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(b) Curare is a poison used by South American Indians when hunting. Curare blocks receptors on the post synaptic membranes of neurones supplying skeletal muscles and prevents breathing.

(i) Explain how curare can stop the process of breathing.

**(2)**

(ii) Acetylcholinesterase is an enzyme found in the gap between two neurones.

Suggest why acetylcholinesterase inhibitors can be used to reverse the effects of curare poisoning.

**(2)**

**(Total for Question 8 = 8 marks)**

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## 6. Question 8 · part a.ii · 3 marks · Working and answer

Working and answer

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# 7. Question 2 · part a.ii · 1 mark

January 2023 Paper 1

Further practice · Attempt part a.ii only.

**2** (a) A human sperm cell has a tail, a mid-piece containing organelles and a head with a nucleus.

(i) Give the function of the tail.

**(1)**

(ii) Which of these, found in the mid-piece, release energy?

**(1)**

- ☒ **A** chromosomes
- ☒ **B** endoplasmic reticulum
- ☒ **C** mitochondria
- ☒ **D** ribosomes

(iii) The process of meiosis is involved in the production of sperm.

Explain the importance of meiosis in sperm production.

**(3)**

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## 7. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

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# 8. Question 1 · part a.i · 3 marks

June 2023 Paper 2R

Further practice · Attempt part a.i only.

**1** (a) The table shows some cell structures and their functions.

(i) Place one tick (✓) in each row to show the function of each cell structure.

**(3)**

| Cell structure | Function | | |
|---|---|---|---|
| | **makes proteins** | **contains DNA** | **allows substances in and out of cells** |
| nucleus | | | |
| ribosomes | | | |
| cell membrane | | | |

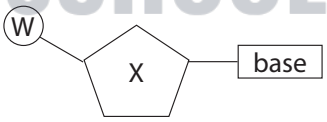
(ii) Give a structure found in bacterial cells.

**(1)**

.....

(b) (i) DNA is made of two chains of nucleotides.

The diagram shows one nucleotide.




Name part W and part X.

**(2)**

W .....

X .....

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## 8. Question 1 · part a.i · 3 marks · Working and answer

Working and answer

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## 9. Question 4 · part a.iii · 1 mark

June 2023 Paper 2

Further practice · Attempt part a.iii only.

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**4** (a) (i) Complete the word equation for one type of respiration.

**(2)**

oxygen + ..... → ..... + .....

(ii) State the type of respiration shown by the equation.

**(1)**

.....

(iii) In which part of the cell does this type of respiration occur?

**(1)**

- ☒ **A** endoplasmic reticulum
- ☒ **B** mitochondrion
- ☒ **C** nucleus
- ☒ **D** ribosome

(iv) Name the molecule used to transfer energy in a cell.

**(1)**

.....

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## 9. Question 4 · part a.iii · 1 mark · Working and answer

Working and answer

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# 10. Question 2 · part a · 1 mark

June 2025 Paper 2R

Further practice · Attempt part a only.

**2** (a) Cells are made up of different components. Each component has a particular function.

Identify the cell component that transports substances inside the cell.

**(1)**

- ☒ **A** cell membrane
- ☒ **B** cytoplasm
- ☒ **C** endoplasmic reticulum
- ☒ **D** ribosomes

(b) Mitosis is a type of cell division.

A student uses a light microscope to view a cell undergoing four stages in mitosis.

The diagrams, W, X, Y and Z, show the student's observations.



W X Y Z

(Source: https://www.pinterest.co.uk/pin/812547957739062607/)

(i) Give the letters to show the correct order of the diagrams in the process of mitosis.

**(1)**

.....

(ii) State one function of mitosis in body cells.

**(1)**

.....

.....

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## 10. Question 2 · part a · 1 mark · Working and answer

Working and answer

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